SCR Hub board (IGVC 2027)

The power, CAN and E-Stop distribution board for Sooner Competitive Robotics' IGVC 2027 robot, with current sensing on every port, designed in Altium.

Features

- 12 V input, 5–8 A total, distributed to seven 6-pin Molex board ports
- Power, CAN bus and the hardware E-Stop line on every port
- INA220 current and voltage sensing on each port
- P-MOSFET reverse-polarity protection (DMP4015SK3Q-13)
- AP63301 synchronous buck for the 3.3 V rail, rated to 32 V input
- Proposed STM32C092 microcontroller with a MAX3051 CAN transceiver
- 8 MHz Abracon ABLS crystal, ±30 ppm, for CAN timing
- SN74AHC244 buffer gives each port its own E-Stop copy
- USB-C debug (CH340C), UART header and test points
- 4-layer board, 100 × 100 mm maximum

Applications

- Power, CAN and E-Stop distribution on Sooner Competitive Robotics' IGVC 2027 robot
- Per-board current and voltage monitoring for the team

Description

The Hub sits between the robot's battery and every other board. It distributes 12 V power, the CAN bus and the hardware E-Stop line to seven 6-pin Molex ports, and it measures the current and voltage on each port so the team can see what every board draws. I redesigned it for 2027 around a proposed STM32C092 microcontroller, P-MOSFET reverse-polarity protection, a 3.3 V buck converter and seven INA220 current sensors on a 4-layer board. Status: schematic complete; PCB layout routed and in final checks.

Team project with Sooner Competitive Robotics (IGVC 2027). My role: schematic, part selection and PCB layout of the Hub.

Key parameters

Parameter	Value
Input	12 V, 5-8 A total
Board ports	7 with current sensing
Board	4-layer 100 x 100 mm max
Status	layout in final checks

> **Data status:** schematic complete, PCB layout routed and in final checks. No values on this page are measured.

Online version: https://saosith.github.io/projects/scr-hub-2027/

Table of contents

1	Introduction	2
2	Requirements	2
3	System overview	2
4	Design	4
5	Design decisions	8
6	Layout	9
7	Project status	13
8	Open issues	14
9	Revision history	14
10	References	14
	Appendix: Bill of materials	14

1 Introduction

The Hub is the board every other board on our robot plugs into. It takes power from the battery and passes three things to each board: 12 V power, the CAN bus and the E-Stop line. For 2027, the team wanted more from it: current and voltage sensing for every board, a temperature reading for fan control, and a measure of how busy the CAN bus is [1]. I started from last year's Hub, reviewed what to keep, and redesigned the power input, the 3.3 V supply, the sensing and the debug access.

Status: schematic complete. **Next:** finish placement and routing, then a team and captain review before ordering. No values on this page are measured.

2 Requirements

- **Distribution:** power, CAN and the E-Stop line to every board through 6-pin Molex connectors [1].
- **Sensing:** current and voltage for every board, plus CAN bus utilization [1]. Temperature sensing for the fan is shared with the Fan Controller board; ownership is still open.
- **Load:** about 5-8 A total for the boards the Hub feeds (team estimate), not counting the motors.
- **Safety:** the E-Stop path must be hardware only, never through firmware [2].
- **Input protection order:** fuse, then a TVS diode in parallel, then series reverse protection [2].
- **Board:** 4-layer boards must stay within 100 x 100 mm [2]. Parts should be surface-mount, hand-solderable, and stocked at DigiKey or JLCPCB/LCSC.

3 System overview

Figure 1 shows the board in three layers. The **power layer** runs from the XT60 through the input protection to a 12 V bus that feeds the seven board ports and the 3.3 V buck. The **data layer** is the CAN bus to every port, plus an I²C bus from the microcontroller to the seven current sensors. The **safety layer** is the E-Stop line, which a buffer copies to every port with no firmware in the path.

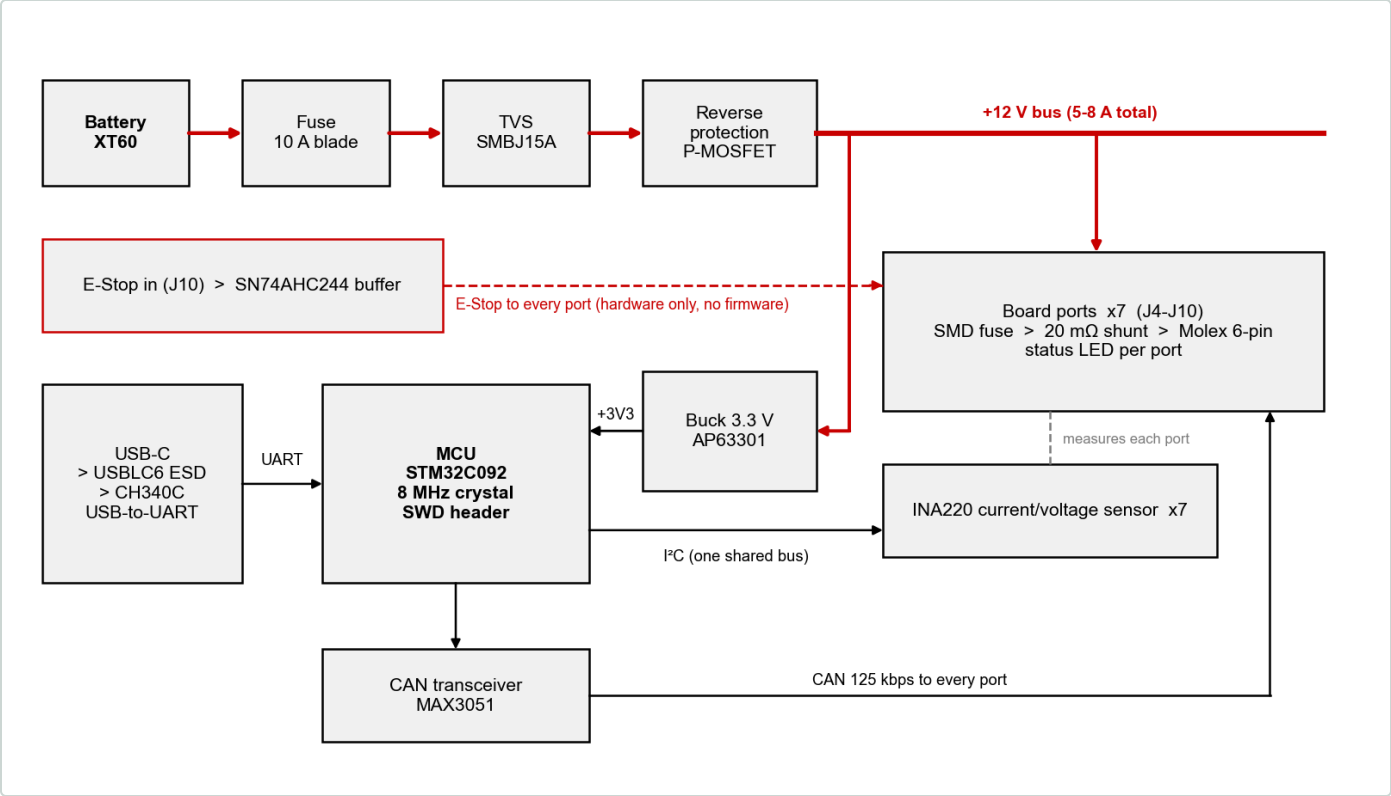


Fig. 1 Hub block diagram. Red lines carry 12 V power; the dashed red line is the hardware E-Stop path.

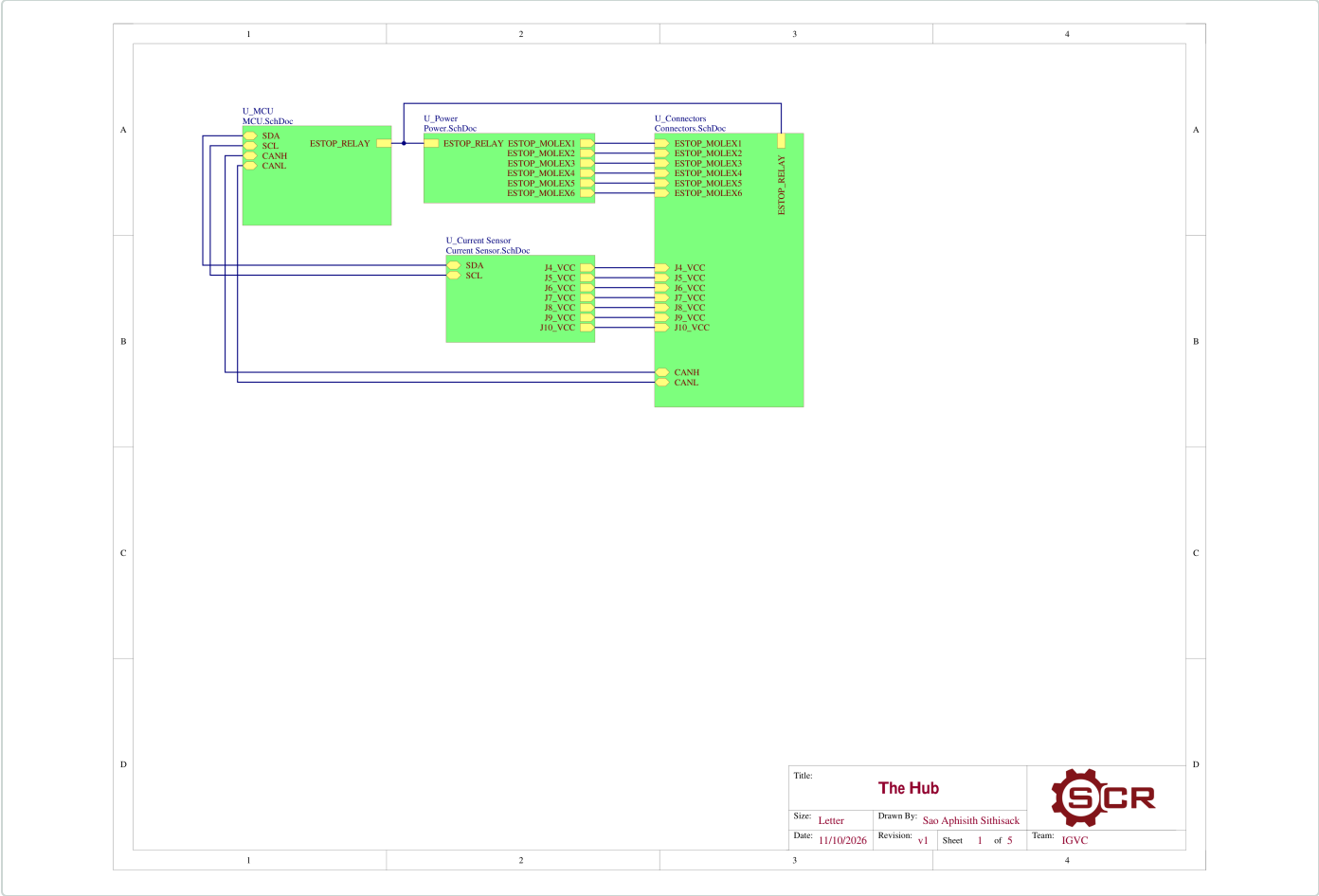


Fig. 2 Schematic top sheet: how the four sheets connect.

4 Design

4.1 Input protection

Battery power enters through an XT60 and passes a 10 A blade fuse in a Keystone 3557-2 holder, then an SMBJ15A TVS diode that clamps voltage spikes to ground. Normally the team uses a series Schottky diode for reverse-polarity protection [2]. At 8 A, a Schottky drops about 0.45 V and dissipates about 3.6 W. I used a P-MOSFET instead: the DMP4015SK3Q-13 has 11 mΩ of on-resistance at V_{GS} = −10 V [3], so $P = I^2 \times R = 8^2 \times 0.011 \approx 0.7$ W.

The drain faces the fuse and the source faces the 12 V bus, so the body diode conducts first and the channel then turns on. If the battery is connected backwards, the gate cannot turn on and the MOSFET blocks. A 10 kΩ resistor pulls the gate to ground, and a 12 V Zener clamps the gate-source voltage within the ±25 V rating. A 100 µF bulk capacitor, a 10 µF ceramic and a 0.1 µF ceramic sit on the bus.

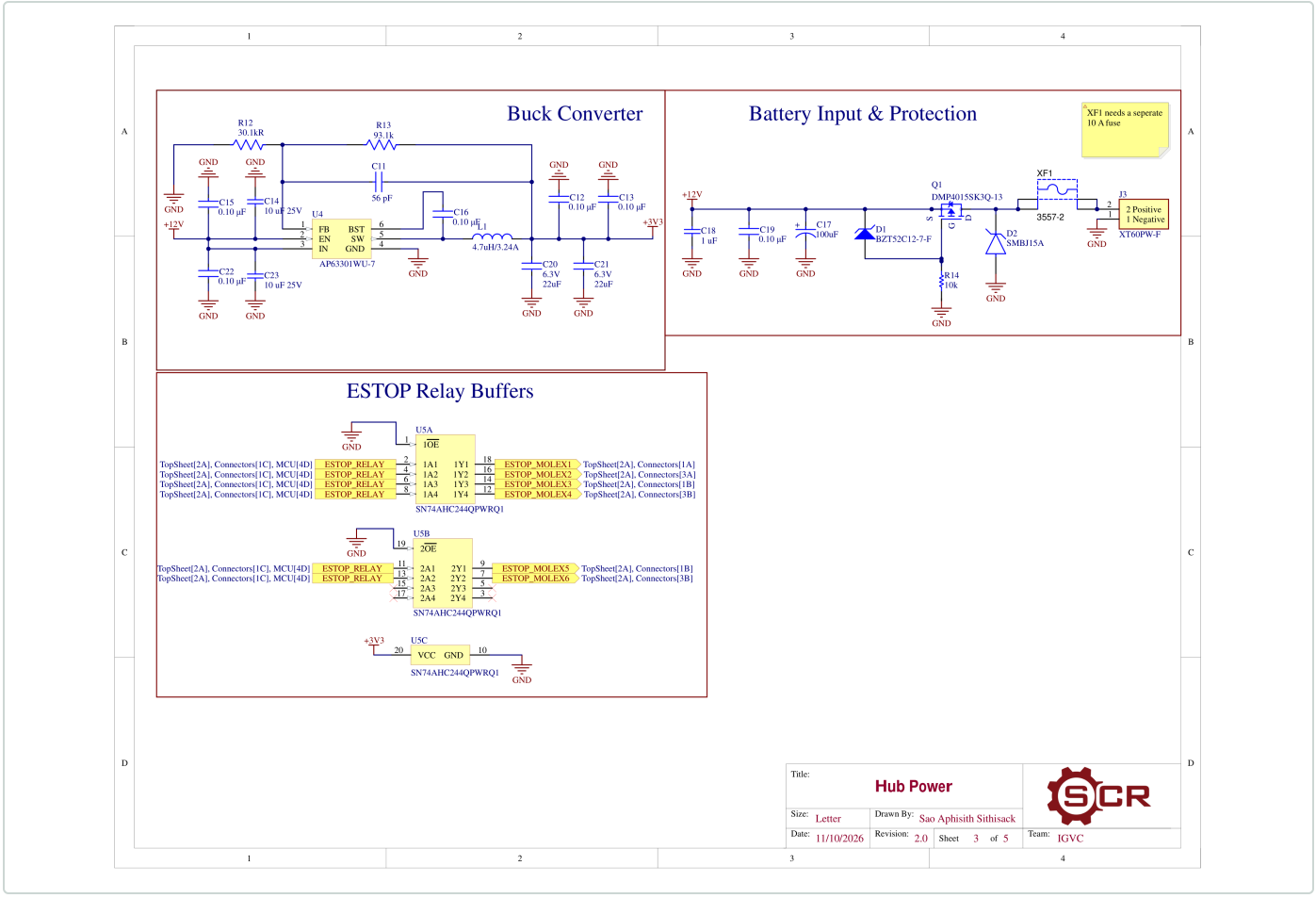


Fig. 3 Power sheet: buck converter (top left), battery input and protection (top right) and E-Stop buffers (bottom).

4.2 3.3 V supply

Last year's Hub used an LDO, which turns the whole 12 V to 3.3 V drop into heat. I replaced it with an AP63301 synchronous buck converter (3.8-32 V input, 3 A, 500 kHz) [4]. The component values follow the datasheet's 3.3 V table: R12 = 93.1 kΩ and R13 = 30.1 kΩ for feedback, 56 pF feed-forward, a 4.7 µH inductor, a 0.1 µF bootstrap capacitor, 10 µF at the input and 2 × 22 µF at the output. Because the converter is synchronous, no catch diode is needed.

4.3 Board ports and current sensing

Each of the seven ports has the same chain: an SMD fuse, a 20 mΩ shunt resistor, then pin 1 of the Molex connector. Pin 6 of each connector drives a status LED through 1 kΩ. An INA220 measures the voltage across the shunt and the port voltage [6]. At 2.5 A the shunt drops 50 mV, inside the INA220's ±80 mV range, and its 10 µV step gives 10 µV / 20 mΩ = 0.5 mA resolution. The sense lines are routed as Kelvin connections from the shunt pads. All seven sensors share one I²C bus, with unique addresses set by strapping A0 and A1.

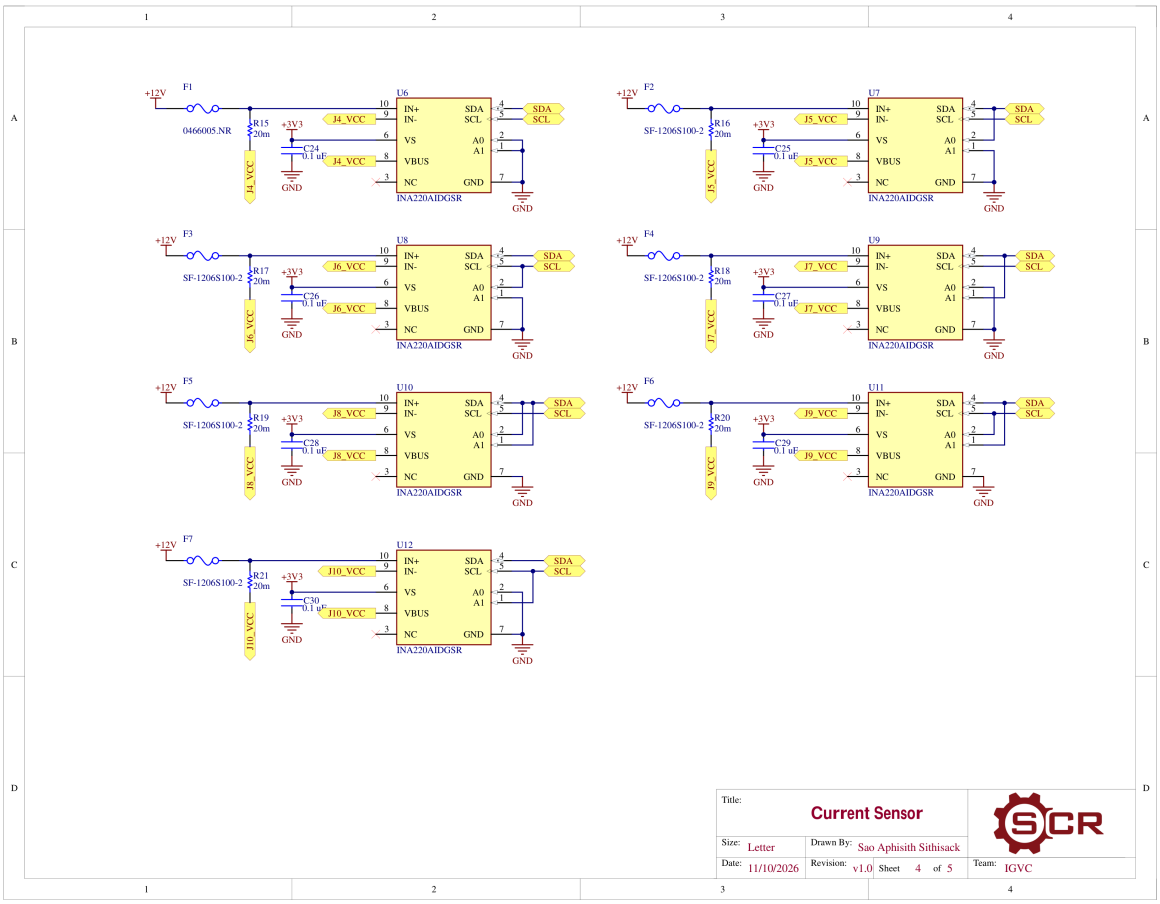


Fig. 4 Current sensor sheet: one fuse, shunt and INA220 per port.

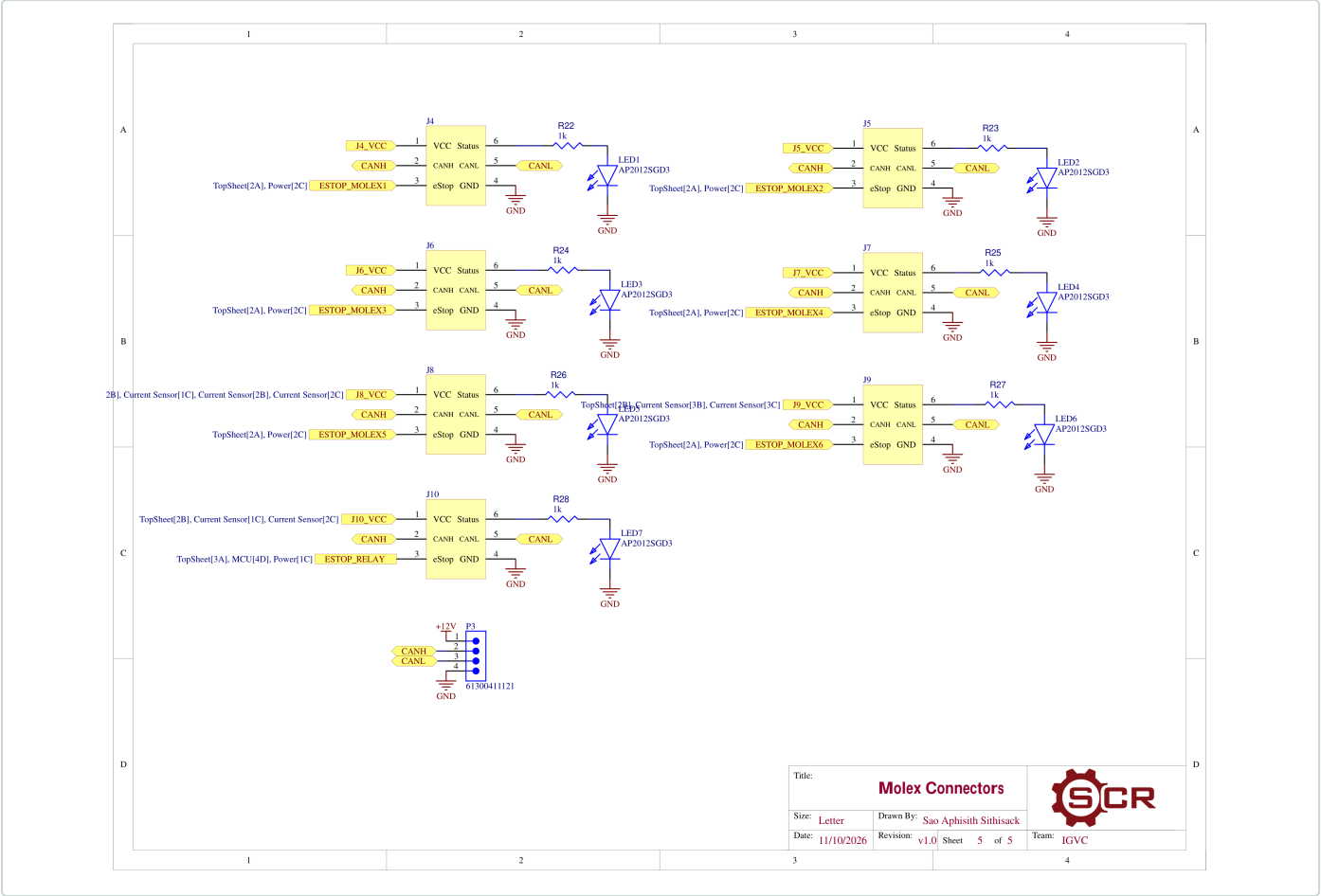


Fig. 5 Connectors sheet: the seven 6-pin Molex ports and status LEDs.

4.4 Microcontroller, clock and CAN

The STM32C092 has a built-in CAN controller (FDCAN), so it needs only a MAX3051 transceiver. This matches last year's Hub and Safety Lights boards, so their firmware can be reused. The internal HSI48 oscillator can drift ±1% over 0-85 °C [5], and two nodes drifting in opposite directions can break CAN timing, so the CAN clock comes from an 8 MHz crystal (Abracon ABLS-8.000MHZ-B4-T). At 125 kbps that gives 8 MHz / 125 kbps = 64 clock cycles per bit. The crystal's load capacitance is C_L = 18 pF, so the load capacitors are $C = 2 \times (C_L - C_{stray}) = 2 \times (18 - 10) = 16$ pF [5]. Bus termination is a 120 Ω resistor with a jumper, so the Hub can be either end of the bus.

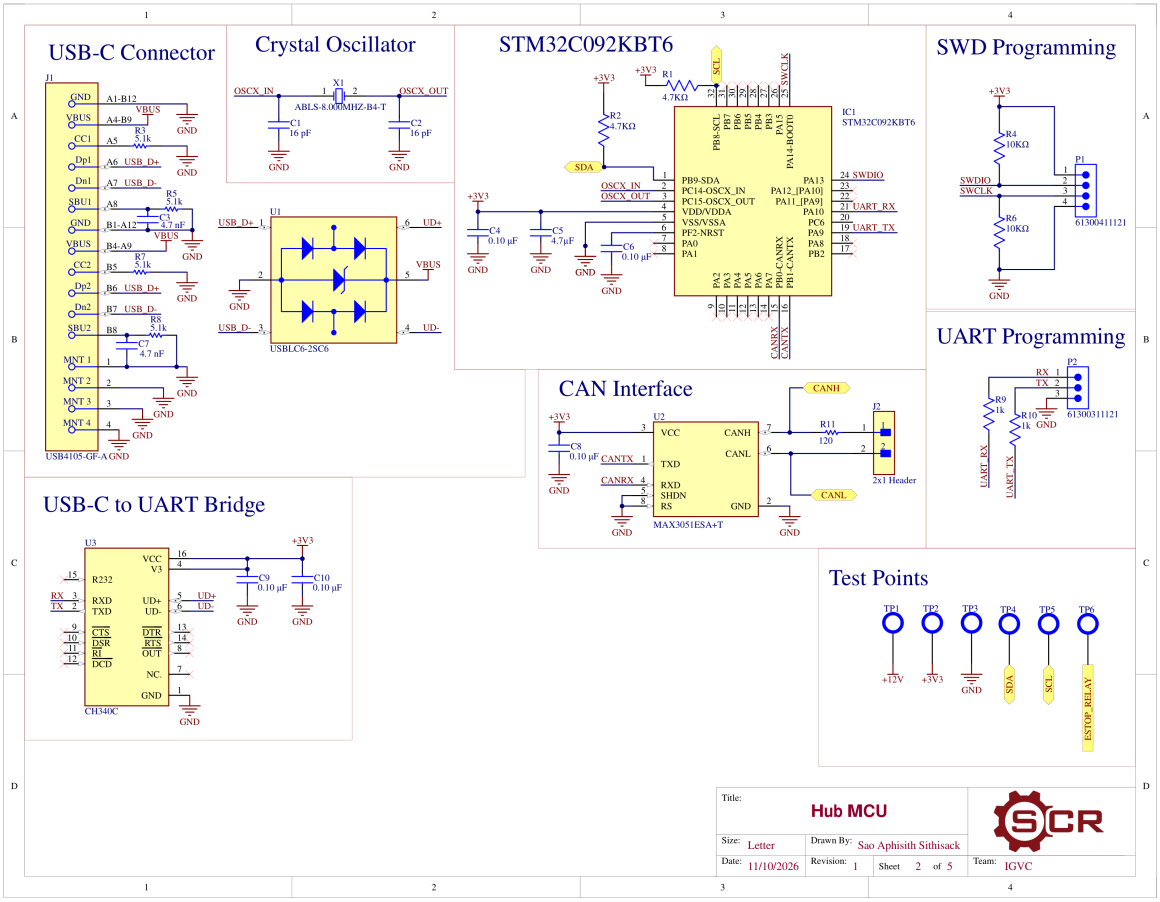


Fig. 6 MCU sheet: microcontroller, crystal, CAN transceiver and USB-to-UART debug.

4.5 E-Stop path

The E-Stop line is active-low 3.3 V logic: high means the robot may run. It comes in from the E-Stop relay board on J10, so J10's E-Stop pin is the input, and an SN74AHC244 buffer drives a separate copy to the other six ports (J4–J9). Firmware never touches this path.

4.6 Debug access

A USB-C connector feeds a USBLC6-2SC6 ESD array and a CH340C USB-to-UART bridge, which connects to the microcontroller's UART through 1 kΩ resistors. A 3-pin UART header sits on the microcontroller side of those resistors, so an external adapter can still be used. Test points cover +12 V, +3.3 V, CANH, CANL, the E-Stop line, SDA, SCL and ground.

5 Design decisions

Decision	Options considered	Choice	Why	Status
Reverse protection	Series Schottky (team guide default), or a P-MOSFET	**P-MOSFET DMP4015SK3Q-13**	At 8 A a Schottky drops about 0.45 V and dissipates about 3.6 W. The MOSFET dissipates about 0.7 W. I chose it for efficiency and less heat.	decided

3.3 V supply	Last year's LDO, or a buck converter	**AP63301 buck**	An LDO's efficiency is at most V_out / V_in = 3.3 / 12 ≈ 28% here. At the Hub's estimated 50–100 mA load, the AP63301 datasheet curve for 12 V to 3.3 V shows about 55–70%, and it accepts up to 32 V.	decided
Microcontroller and CAN	Pico + MCP2515 (team standard), or STM32C092 + MAX3051	**STM32C092 + MAX3051**	One chip handles CAN, it matches the 2026 Hub and Safety Lights, and their firmware can be reused.	open
MCU clock	Internal HSI48 oscillator, or an 8 MHz crystal	**8 MHz crystal, Abracon ABLS (16 pF load caps)**	HSI48 can drift ±1% over temperature, and two nodes drifting apart can break CAN timing. The ABLS is ±30 ppm and has a footprint in the library.	decided
I²C buses	One shared bus, or two buses for fault isolation	**One bus**	Standard for about 8 devices; firmware timeouts and bus recovery handle a stuck device.	open
Debug access	None, test points only, UART header, or on-board USB-C	**USB-C (CH340C) + UART header + test points**	One cable for debugging, with the header as a backup and for probing.	decided
Layer count	2-layer, or 4-layer	**4-layer**	A solid ground plane makes the buck, USB and CAN layout forgiving for a first board and frees routing space.	decided
Main fuse rating	10 A, or 15 A	**10 A for now**	Depends on whether the NUC computer is powered through the Hub (about 2 A more).	open

6 Layout

The board is 4-layer: signals on L1 and L4, a solid ground plane on L2, and a solid +12 V plane on L3, on PCBWay's 1.6 mm stack-up [7]. The +3.3 V rail carries well under 1 A, so I routed it as wide traces on the signal layers and kept L3 whole for the 8-10 A that feeds the seven port fuses. The design rules keep a margin over PCBWay's minimums: 0.2 mm clearance, 0.3 mm / 0.6 mm vias, 0.5 mm copper to the board edge, and wider minimum widths for the power nets (1.0 mm for port power, 1.5 mm for +12 V). USB D+/D− is routed as a 90 Ω differential pair.

The Molex ports sit around the edges with their port blocks (fuse, shunt, INA220) right behind them. The XT60 input, fuse holder and P-MOSFET are on the right, the microcontroller is in the center with the crystal at its pins, the USB-C connector is on the bottom edge, and the buck converter is in the bottom-right corner, away from the crystal and the USB lines. The 10 A input path is drawn as wide pours on the outer layers.

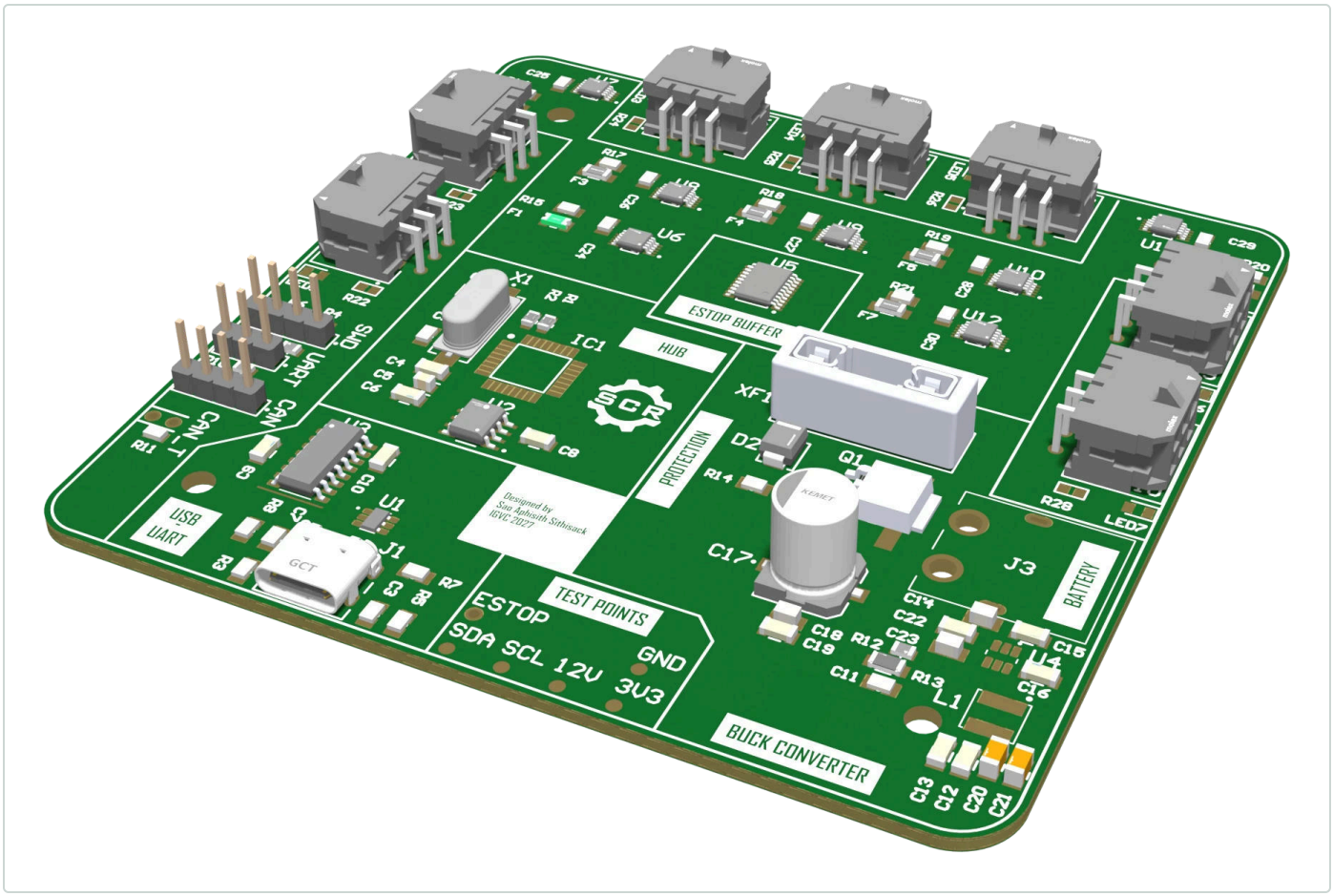


Fig. 7 3D render of the routed board.

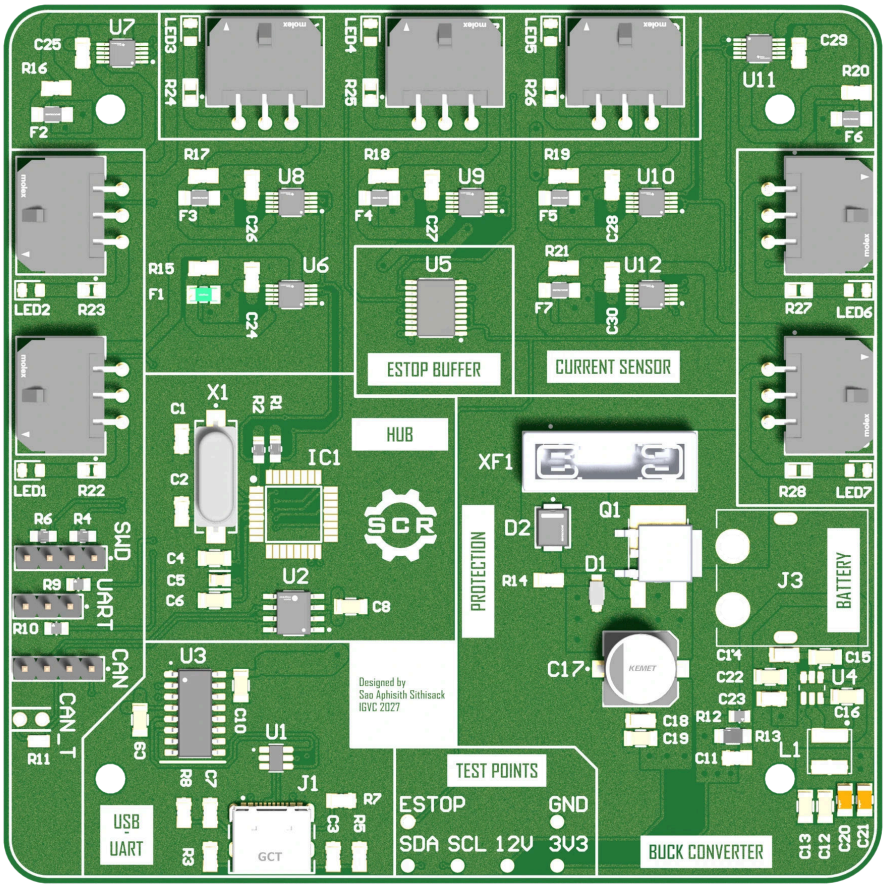


Fig. 8 Top view. Each circuit block is outlined and labeled on the silkscreen.

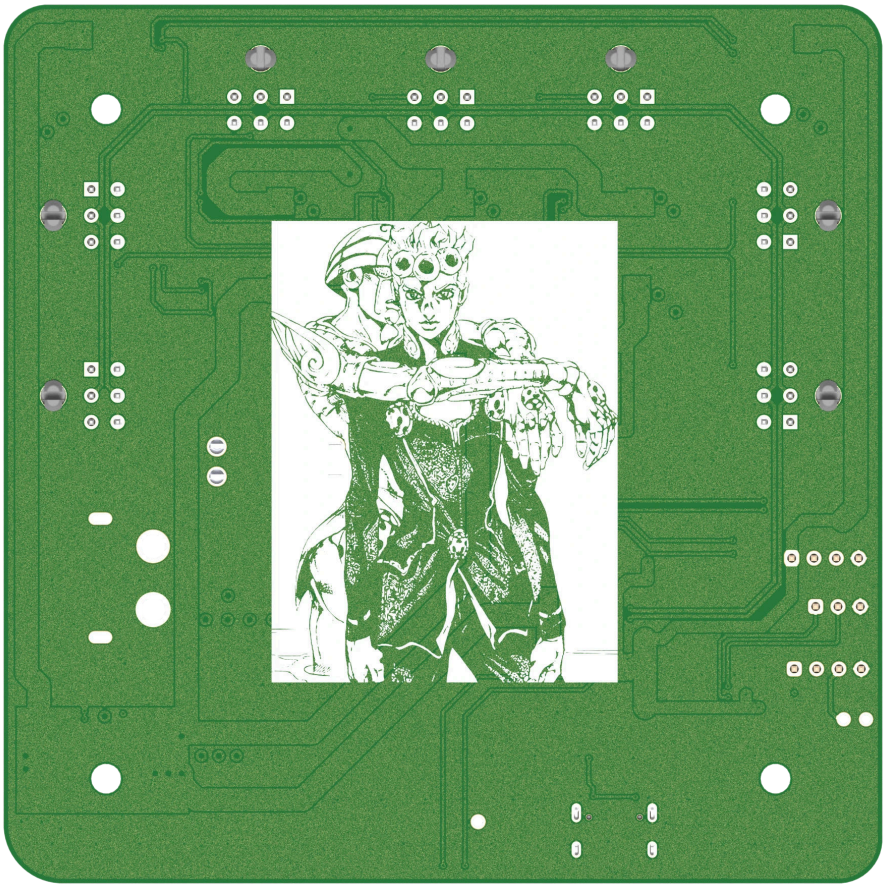


Fig. 9 Bottom view: bottom-layer routing and silkscreen art.

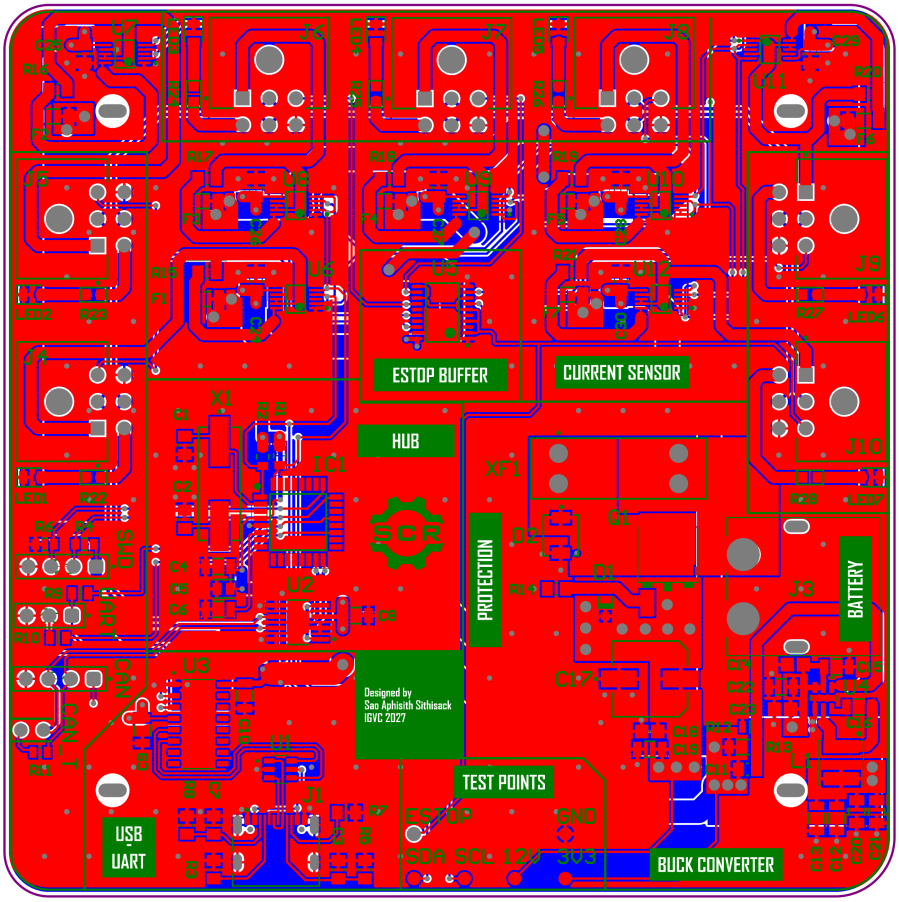


Fig. 10 Altium PCB print: top copper in red, bottom copper in blue.

7 Project status

Sections for simulation and test results are added as each stage is completed.

Stage	Status	Plan
Requirements and architecture	✓ Complete	From the 2027 task document and a review of last year's Hub.
Schematic	✓ Complete	Five sheets, in the Altium PDF. Small fixes listed under Open issues.
Design rules and stack-up	✓ Complete	PCBWay 4-layer 1.6 mm stack-up; rules with margin over PCBWay minimums.
Placement and routing	▶ In progress	Routed. Final fixes before review: buck feedback placement, ground vias, part numbers.
Design review	○ Planned	Team and captain review before any order.
Fabrication	○ Planned	PCBWay, 4-layer 1.6 mm, after captain approval.
Firmware	○ Planned	Start from the 2026 Safety Lights code (same STM32C092); report port currents over CAN.
Bring-up and test	○ Planned	Rail voltages, reverse-polarity test, current readings against a bench meter, CAN and E-Stop checks.

8 Open issues

- **Main fuse rating.** 10 A, or 15 A if the NUC is powered through the Hub. Waiting on the team's power plan.
- **Port fuse ratings.** Each board's maximum current is still needed to size its fuse. Port 1 uses a Littelfuse 0466005.NR while ports 2-7 use SF-1206S100-2; these should be reconciled.
- **Spare E-Stop buffer inputs.** Pins 15 and 17 of the SN74AHC244 are marked no-connect; tie them to ground.
- **Port surge protection.** The ports have fuses but no TVS diodes; decide whether each port needs one.
- **Overlap with the Sensor Board.** Both boards measure the main current and temperature; the team needs to decide which board owns what.
- **Approvals.** The STM32C092, the single I²C bus, the P-MOSFET and the PCBWay order all need captain approval.

9 Revision history

Rev	Date	What changed
A	2026-09-24	Project started; requirements collected from the team documents.
A	2026-10-01	Schematic started from last year's Hub; crystal and buck converter chosen.
A	2026-10-04	Reverse protection changed from a Schottky to a P-MOSFET.
A	2026-10-05	Buck changed to the AP63301; on-board USB-C to UART added.
A	2026-10-08	4-layer board chosen; PCBWay stack-up and design rules set (10-09 to 10-10).
A	2026-10-12	Placement and routing in progress; main input path drawn as pours.
A	2026-11-10	Layout routed; schematic PDF updated; 3D renders and schematic figures added.

10 References

1. Sooner Competitive Robotics, IGVC 2027 Electrical Requirements and Tasks
2. Sooner Competitive Robotics, Board Design Guide, Rev 0.5
3. Diodes Inc., DMP4015SK3 P-channel MOSFET datasheet
4. Diodes Inc., AP63300/AP63301 synchronous buck converter datasheet
5. STMicroelectronics, STM32C092 datasheet, and AN2867 oscillator design guide
6. Texas Instruments, INA220 current and power monitor datasheet
7. PCBWay, 4-layer 1.6 mm standard stack-up

Appendix: Bill of materials

Ref	Part	Value	Package	Source
J3	Amass XT60PW-F battery connector	—	XT60 right-angle	TBD
XF1	Keystone 3557-2 fuse holder (10 A blade fuse)	—	ATO clip	LCSC C352820
D2	SMBJ15A input TVS	—	SMB	TBD
Q1	Diodes Inc. DMP4015SK3Q-13 P-MOSFET	—	TO-252	LCSC C461089

D1	Diodes Inc. BZT52C12-7-F gate Zener	12 V	SOD-123	LCSC C124196
U4	Diodes Inc. AP63301WU-7 buck converter	3.3 V out	TSOT-26	TBD
IC1	ST STM32C092KBT6 microcontroller	—	LQFP-32	TBD
X1	Abracon ABLS-8.000MHZ-B4-T crystal	8 MHz	HC-49/US SMD	LCSC C596838
U2	Maxim MAX3051ESA+T CAN transceiver	—	SOIC-8	TBD
F1-F7	Port fuses: Littelfuse 0466005.NR (F1), SF-1206S100-2 (F2-F7)	—	1206	TBD
R15-R21	Yageo RL1206FR-070R02L port shunt	20 mΩ	1206	LCSC C513558
U6-U12	TI INA220AIDGSR current sensor	—	VSSOP-10	TBD
U5	TI SN74AHC244QPWRQ1 E-Stop buffer	—	TSSOP-20	TBD
LED1-LED7	AP2012SGD3 port status LED	—	0805	TBD
J1	GCT USB4105-GF-A USB-C receptacle	—	SMD	LCSC C3020560
U1	ST USBLC6-2SC6 USB ESD array	—	SOT-23-6	LCSC C7519
U3	WCH CH340C USB-to-UART bridge	—	SOIC-16	LCSC C84681
J4-J10	6-pin Molex board connectors	—	—	TBD